

Terminating Pythagorean Trees and the Periodic Zeta of the Romik Map

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Abstract

We prove a convention-complete split theorem for the Romik three-branch map. Reduced rational parameters with opposite parity terminate at $1/2$ and form the unique (a odd, b even) Barning tree rooted at $(3, 4, 5)$; leg swap is a separate orientation terminating at $1/3$. On the invariant irrational set, every length- n word except the pure 1 and pure 3 parabolic words has one quadratic-irrational fixed point. Hence $\# \text{Fix}(T^n) = 3^n - 2$, Möbius inversion gives every primitive count, and the source Artin–Mazur zeta is $(1 - z)^2 / (1 - 3z)$. Word matrices also give exact orientation and multipliers. A 9,840-word finite ledger audits the formulas. The terminating arithmetic states and periodic states are disjoint, so no rational-prime orbit law or target-zeta claim follows.

Revision certificate. Quadratic periodic atlas, primitive counts, and source zeta.

1 Frozen map and orientations

For $0 < t < 1$, put

$$D(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right), \quad T(t) = \begin{cases} t/(1-2t), & 0 < t < 1/3, \\ 1/t - 2, & 1/3 < t < 1/2, \\ 2 - 1/t, & 1/2 < t < 1. \end{cases}$$

The branch images are open. We declare $1/2$ the primary terminal for the odd-first/even-second triple $(3, 4, 5)$, while $1/3$ is the terminal for the leg-swapped $(4, 3, 5)$ orientation. Neither endpoint belongs to the periodic space $X = \{t \in (0, 1) : t \notin \mathbb{Q}\}$. The inverse branches are

$$F_1(t) = \frac{t}{1+2t}, \quad F_2(t) = \frac{1}{2+t}, \quad F_3(t) = \frac{1}{2-t},$$

with images $(0, 1/3), (1/3, 1/2), (1/2, 1)$.

Theorem 1. *The three Barning matrices*

$$M_1 = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}$$

give every primitive Pythagorean triple with first leg odd and second leg even exactly once as $M_{w_1} \cdots M_{w_n}(3, 4, 5)^t$ for a possibly empty finite word w ; the empty word owns the root $(3, 4, 5)$. For every $n \geq 1$, every word $w \in \{1, 2, 3\}^n$ except $1^n, 3^n$ has one fixed point t_w in its open inverse cylinder, and these are exactly $\text{Fix}(T^n|X)$. They are quadratic irrational, and

$$\# \text{Fix}(T^n|X) = 3^n - 2, \quad E_n = \sum_{d|n} \mu(d)(3^{n/d} - 2), \quad \pi_n = E_n/n,$$

where E_n counts exact-period points and π_n oriented cycles. Moreover

$$\zeta_T(z) = \exp\left(\sum_{n \geq 1} \frac{3^n - 2}{n} z^n\right) = \frac{(1-z)^2}{1-3z}.$$

If $F_w(t) = (at + b)/(ct + d)$, then t_w solves $ct^2 + (d-a)t - b = 0$, $\det F_w = (-1)^{\#_2(w)}$, and

$$|(T^n)'(t_w)| = \rho(F_w)^2, \quad \rho(F_w) = \frac{\text{tr } F_w + \sqrt{(\text{tr } F_w)^2 - 4 \det F_w}}{2}.$$

2 Rational descent and unique tree

Write $t = n/m$ in lowest terms. If $m > n > 0$ have opposite parity, then

$$(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$$

is primitive with a odd and b even, and $D(t) = (a/c, b/c)$. Conversely, for such a primitive triple, $(c + a)/2$ and $(c - a)/2$ are coprime and have square product $(b/2)^2$; hence they are m^2, n^2 , proving the unique Euclid parameters.

The inverse branches replace (m, n) by

$$(m + 2n, n), \quad (2m + n, m), \quad (2m - n, m).$$

Substitution in the three quadratic formulas gives exactly multiplication by M_1, M_2, M_3 . Coprimality and opposite parity persist. In the forward direction the three branches give

$$\frac{n}{m - 2n}, \quad \frac{m - 2n}{n}, \quad \frac{2n - m}{n};$$

each new denominator is smaller than m . Opposite parity excludes the $1/3$ terminal, so descent ends at $1/2$. Disjoint branch intervals make each removed digit unique. This proves the primary tree; leg swap remains a separate mirror orientation.

3 Periodic-word lemma and zeta

Lemma 1. *The words 1^n and 3^n have only the boundary fixed points 0 and 1. Every other F_w is a strict contraction of $[0, 1]$ into an interior subinterval and has one quadratic-irrational fixed point in its cylinder.*

Proof. The derivative magnitudes of the inverse branches are $(1 + 2t)^{-2}, (2 + t)^{-2}, (2 - t)^{-2}$. Equality with one is possible only for F_1 at zero or F_3 at one; $|F'_2| \leq 1/4$. Thus a nonpure composition maps $[0, 1]$ into $(0, 1)$ and has derivative supremum below one. The contraction theorem gives one fixed point. The pure powers satisfy $F_{1^n}(t) = t/(1 + 2nt)$ and, by reflection, the analogous formula at one.

For $F_w = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the fixed equation is $ct^2 + (d - a)t - b = 0$. Its derivative is $\det(F_w)/(ct + d)^2$. Every branch denominator is positive on $[0, 1]$; the denominator cocycle therefore gives $ct_w + d > 0$. At the attracting point this is an eigenvalue $\lambda > 1$, while the other is $\det(F_w)/\lambda$. Thus the trace is $\lambda + \det(F_w)/\lambda > 0$, the displayed positive square-root formula is exactly $\rho(F_w) = \lambda$, and the matrix discriminant is positive. If it were square, determinant 1 would factor $\text{tr}^2 - r^2 = 4$ and force the parabolic trace two; determinant -1 would factor $r^2 - \text{tr}^2 = 4$ and force trace zero. Both contradict strict contraction. The fixed point is quadratic irrational. $\square$

Equal-length cylinders are disjoint, so their fixed points are distinct. Every point of $\text{Fix}(T^n|X)$ has one itinerary and is fixed by its inverse word. Removing the two parabolic words gives $3^n - 2$. Since fixed points partition by exact period, Möbius inversion gives E_n and division by n gives π_n . Finally

$$\sum_{n \geq 1} (3^n - 2)z^n/n = -\log(1 - 3z) + 2\log(1 - z),$$

which proves the source zeta. The eigenvector identity $F_w(t_w, 1)^t = (ct_w + d)(t_w, 1)^t$ and the derivative formula give the stated multiplier. Matrix powers make logarithmic instability additive under word repetition. A word with an odd number of digit 2 reverses orientation and is not silently treated as a PSL_2 element.

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